POL-SAR Image Classification Based on Modified Stacked Autoencoder Network and Data Distribution

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***Abstract*— This article proposes a novel autoencoder (AE) network based on the distribution of polarimetric synthetic aperture radar (POL-SAR) data matrix, called a mixture autoencoder (MAE). Through a detailed analysis of the data distribution POL-SAR data matrix, a normalization method is also presented in succession. The proposed MAE defines the data error term in the loss function according to the data distribution. It can be regarded as a process of unsupervised feature extraction designed specifically for POL-SAR data matrix. Then, a softmax classifier is trained with the help of data features and the corresponding label information. Next, a stacked MAE (SMAE) network is reasonably constructed by considering the data distribution among different layers. Finally, this article also presents a classification network through discarding the decoder process of the proposed SMAE and connecting with a softmax classifier. The SMAE is trained layer by layer using the unlabeled data. The softmax classifier is also trained with a small number of labeled pixels. With parameters obtained from the above-mentioned procedures as the initial parameters, the whole classification network is trained by the labeled pixels to get a well-trained model, which is used for predicting the corresponding label of the pixel in the data set. Three real POL-SAR data sets, including the AIR-SAR L-band data of Flevoland, The Netherlands, are used in the experiments. Compared with one classical algorithm and two related models with the similar structure, both the proposed methods show improvements in overall accuracy and efficiency as well as possess better adaptability of the parameter and preferable consistency with the classification performance.**

***Index Terms*— Autoencoder (AE) network, classification network, coherency matrix, image classification, polarimetric synthetic aperture radar (POL-SAR) data matrix, softmax classifier, Wishart autoencoder, Wishart distribution.**

I. Introduction

FOR recent decades, polarimetric synthetic aperture radar (POL-SAR) has been one of the most widely used and advanced sensors in the field of remote sensing. It emits and/or receives electromagnetic energy in the direction of vertical and horizontal polarization, which makes much richer characteristics of the observed land cover and targets. Similar to the synthetic aperture radar, the POL-SAR also has a good ability to adapt to the situation of day and night and all weather. It has been highly applied in many domains, for example, military and agriculture [1]. As one of the most important and fundamental applications of remote sensing, POL-SAR image classification has been studied by many scholars for a long time. By definition, the task of POL-SAR image classification is assigning one class label to each pixel. A great variety of methods for POL-SAR image classification have been put forward and developed over the past several decades, which mainly divided into three categories: statistical distribution of POL-SAR data [2]–[5], polarization decomposition [6]–[9], and machine learning algorithms [10]–[13].

Some researchers employed the statistical distribution of original POL-SAR data to solve the problem of classification. As it is known to all, both the covariance matrix and the coherency matrix of POL-SAR data are assumed to subject to a complex Wishart distribution [4], [14]. Wishart distance is proposed by Lee *et al.* [2], [3] based on the complex Wishart distribution of covariance matrix and coherency matrix. It has been one of the most outstanding achievements in the usage of the statistical distribution of POL-SAR data. The Wishart classifier classifies each pixel according to the distance between the pixel and the center of each category. The performance of this classifier is determined by the quality of center of each category.

In view of the mechanism of polarization scattering, a number of polarization decomposition methods have been put forward to classify the POL-SAR image, such as entropy/alpha decomposition [7], Freeman decomposition [8], and Krogager decomposition [9]. Among all these decomposition methods, the entropy/alpha decomposition proposed by Cloude and Pottier [7] is one of the most extensively used method. The coherency matrix is decomposed into three components of different scattering mechanisms. The pixels in the POL-SAR image are classified according to the parameters obtained by the H/α polarization decomposition. Although this method can only classify eight kinds of land objects, it is still popular for its simplicity and clearly physical meanings [15]. It is obvious that entropy/alpha decomposition is an unsupervised classification method. In addition, the statistical property of POL-SAR image is utilized together with its scattering mechanism. Lee [3] used the result of entropy/alpha decomposition to

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initialize the classification of POL-SAR image. Then, repeatedly calculating the mean of pixels for each category and making use of the Wishart distance to reclassify every pixel in the POL-SAR image until the termination criterion was satisfied. As described earlier, these polarization decomposition methods highly rely on the complex analysis of the POL-SAR data, which is difficult and time-consuming. In addition, the abovementioned methods of utilizing statistical information directly or indirectly give part of the labeled information as the initial classification center of each category. Meanwhile, none of them attempt to exploit the unlabeled pixels in the classification of POL-SAR image. In practice, it is labor-intensive and time consuming to label the POL-SAR image pixel by pixel. As a result, only a small number of the labeled pixels are available, and most of the pixels are unlabeled in many cases. Therefore, it will be exciting if numerous unlabeled pixels are properly used.

As one of the most important and effective branches of artificial intelligence, the method of machine learning is also applied in the classification of POL-SAR image, such as support vector machine (SVM) [16], [17], neural networks (NNs) [11], [18]–[21], and AdaBoost [22]. The procedure of POL-SAR image classification with machine learning can be divided into two stages: features learning and classifier training. The first stage acts as a feature extractor, which does not need to consider whether the data are labeled or not. The second stage is choosing a classifier and training it with the help of a few labeled pixels. Although machine learning methods have strong classification ability to achieve good result, they depend heavily on the features of pixels. It requires professional analysis and plenty of experiments to decide what features should be extracted and how to obtain good features.

As one of the most popular deep NNs, deep belief network (DBN) was not widely used until 2006 when Hinton *et al.* [23] proposed an efficient greedy learning algorithm and used pretraining method to alleviate the problem of local optimal solution, which made it possible for the NN to contain more hidden layers. NN has "depth" in its true sense, which opens the upsurge of deep learning. It performs well in many fields, including the speech recognition [24], image recognition [25], and natural language processing [26]. The autoencoder (AE) network is a typical NN, which is used to produce the output that approaches the input. Since the information of interest is contained with the activation of hidden units [10], it can also be used to extract the features of samples. The stacked AE (SAE) is an NN, which is made up of multiple AE NNs. The output of the first layer of AE NN serves as the input of the second layer of AE NN. SAE is a deep NN that uses the abovementioned unsupervised pretraining. The convolutional NN (CNN) has got significant achievement in image classification [27], [28], whose first stage acts as a feature extractor, in which the features mainly obtained through a convolution operation, and the second stage is a multilayer perceptual classifier. The CNN is a method of supervised feature learning. Compared with the AE, the superiority of the CNN is the usage of spatial context when classifying the pixel-based image and it is well known that the spatial information plays an important role in all image classifications. The introduction of spatial information undoubtedly leads to an increase in the amount of data, which raises higher requirements for memory capacity. In the meantime, the operation of convolution on a large number of image data is extraordinary time-consuming. Combining the advantages of CNN and AE, Masci *et al.* [29] proposed a novel convolution AE (CAE) for unsupervised feature learning. The CAE is trained using conventional online gradient descent without additional regularization terms, and the max-pooling layer is essential to learn biologically plausible features consistent with those found by previous approaches. Similar to the previous two classification methods, CNN and CAE can be cascaded to form deep convolution NN and deep convolution autoencoder network, respectively.

The machine learning algorithms mentioned earlier have been directly employed for the classification of POL-SAR image without taking data distribution into consideration. For example, LV modeled the POL-SAR data by the general DBN [30]. CAE has been successfully used in the high-resolution SAR image classification [10]. In recent years, more and more attention has been paid to integrating the distribution characteristics of POL-SAR data into machine learning algorithms. Taking complex Wishart distribution of the coherency matrix of POL-SAR into account, Xie *et al.* [31] proposed a novel features' extraction model during the construction of AE network, which named Wishart AE (WAE). It combines Wishart distance with the backpropagation (BP) algorithm through the training procedure of AE. Generally, features learned by WAE should be more suitable for the classification task. Guo *et al.* [32] gave rudimentary research and proposed the Wishart restricted Boltzmann machine (WRBM) by considering the Wishart distribution of covariance matrix. Then, a more rigorous and reasonable study was made by Liu [33] *et al.*, in which the modified restricted Boltzmann machine (RBM) named Wishart–Bernoulli RBM (WBRBM) is proposed and the corresponding DBN is named W-DBN. The energy function of WBRBM is defined by taking the data distribution into consideration, which is much more stringent mathematically. Instead of one visible node, all of them correspond to a POL-SAR pixel. The WBRBM is set as the lowest layer of the W-DBN and others are all binary RBMs, which is a much more plausible explanation for data distribution among different layers.

At the same time, it is more popular to combine other kinds of methods with the machine learning algorithms to complete the task of POL-SAR image classification. For example, local spatial information was introduced into a stacked sparse AE to learn the deep spatial sparse features automatically for the first time in [34]. A new POL-SAR image classification method has been proposed [35], in which superpixels are utilized to integrate contextual information of neighborhood and a multilayer AE network is used to learn the features for distinguishing the category of each pixel. To exploit the polarimetric information in the POL-SAR image, H-A-α decomposition was integrated with fully convolutional networks [36]. The features obtained from an RBM with patches, randomly selected from each dimension in the 9-D data, and the elements of coherent matrix

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are combined as the input of a three-layer DBN for POL-SAR image classification [37].

The original AE uses the Frobenius norm to measure the similarity of the input matrix and the output matrix. This similarity measurement method is suited for natural images that obey the Gaussian distribution. However, for pixels of POL-SAR images, they do not obey the Gaussian distribution. Therefore, it is inappropriate to directly apply a machine learning algorithm to complete the task of POL-SAR image classification. In addition, there have been many attempts to integrate the distribution of different forms of POL-SAR data into machine learning models. For example, the complex Wishart distribution of POL-SAR coherence matrix is considered into the construction of data error in the loss function of AE network. The energy function of RBM network is modified by the data distribution of different rows in the POL-SAR matrix. In a word, the distribution of data is playing a more and more important role in designing machine learning model for POL-SAR image classification. Hence, taking the distribution of POL-SAR data matrix into account, a novel feature learning network is given and called the mixture AE (MAE), in which Wishart distance and Euclidean distance are applied to evaluate the data error between the input and the output in the cost function during the process of using the AE network to extract the features of POL-SAR data. In Section II, we first make a further study on the composition and distribution of POL-SAR data matrix. To be specific, all the pixels are rearranged to constitute a POL-SAR data matrix by converting the complex coherency matrix into a 9-D vector. As a matter of fact, the first three rows of the POL-SAR data matrix are considered to be subordinate to Gamma distribution, while the rest rows are approximated with the Gaussian distribution. Therefore, the POL-SAR data matrix is considered to follow a mixture distribution of the Gamma distribution and the Gaussian distribution. Then, a normalization method is proposed by being divided by the variance or standard deviation corresponding to each dimension. According to its distribution, the POL-SAR data matrix can be divided into two parts. Different strategies can be taken to measure the error of data during the process of AE. The errors of first three rows of data matrix can be measured through a simplified Wishart distance, while the data in the remaining rows can still be regarded as a matrix, of which the process method is similar to that of standard AE. It is obvious that the MAE is an unsupervised feature learning process, which does not need the label information. Moreover, it can extract more appropriate and effective features for the task of POL-SAR classification. After feature extraction, a classifier is connected to execute the procedure of classification.

In order to acquire the higher level features of the data for better completing the task of classification of the POL-SAR image, a stacked MAE (SMAE) network is also given based on the proposed MAE. It is not a simple cascade of several MAEs. However, the proposed MAE is merely employed at the first layer of the network and the ordinary AE is set at the rest layers, which fully considered the data distribution of different layers. The SMAE adopts a deep network structure to extract the features of high-dimensional space. Furthermore, it is trained layer by layer with the unlabeled data, and then, a softmax classifier is trained with the help of a few labeled pixels. By discarding the decoding process of the SMAE and integrating with the softmax classifier, a multilayer network for POL-SAR image classification is also presented in this article. Using parameters for the encoding stage of SMAE and the softmax classifier parameters as the initial parameters, the whole classification network is trained with the labeled pixels to get the best parameters.

The rest of this article is organized as follows. The analysis and preprocessing of POL-SAR data are introduced in Section II. Section III presents the details of the proposed MAE methods and the SMAE classification network. The experimental analysis and results are shown in Section IV to verify the effectiveness of our proposed methods. Finally, the conclusion and future work are discussed in Section V.

II. Data Analysis and Preprocessing

The Sinclair scattering matrix [1]

$$S = \begin{bmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{bmatrix} \tag{1}$$

can be used to briefly describe a single pixel in the POL-SAR image, where S_{hv} is the scattering element of horizontal transmitting and vertical receiving polarization. The other elements are defined by the similar way.

According to the reciprocity theorem, the polarimetric scattering matrix based on the Pauli basis can be vectorized as

$$k_{3p} = \frac{1}{\sqrt{2}} \left[S_{hh} + S_{vv}, S_{hh} - S_{vv}, 2S_{hv} \right]^T. \tag{2}$$

Therefore, the coherence matrix can be calculated by the following formula:

$$\begin{aligned} [\mathrm{T}] = k_{3p} k_{3p}^{*T} &= \begin{bmatrix} |e^2| & ef^* & eg^* \\ fe^* & |f^2| & fg^* \\ ge^* & gf^* & |g^2| \end{bmatrix} \\ &= \begin{bmatrix} 2A_0 & \text{C-jD} & H + jG \\ C + jD & B_0 + B & E + jF \\ H - jG & \text{E-jF} & B_0 - B \end{bmatrix} \end{aligned} \tag{3}$$

where e, f, and g are the corresponding elements in the vector k_{3p}, and superscript T and superscript $*$ denote a transposition operator and a conjugate operator, respectively. Because the data form of each element in the Sinclair scattering matrix is complex number, the elements, which used to constitute the vector k_{3p}, are also complex number. Therefore, the last step of (3) can be obtained by the form of a general complex matrix after further numerical calculation.

Based on the scattering matrix, the corresponding coherency matrix contains fully polarimetric information as well. Moreover, the coherency matrix retains the second-order statistics of polarimetric information. Simultaneously, due to the inherent speckle characteristic of POL-SAR, the signal-to-noise ratio (SNR) of the data is reduced, which brings great difficulties to the subsequent analysis and processing. Thus, multilook processing is usually one of the basic methods for speckle suppression of POL-SAR images [38]. It has been

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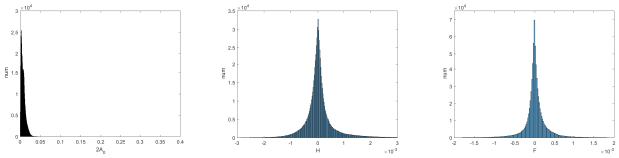
proven in the literature [14] that multilook coherency matrix follows the complex Wishart distribution.

In this article, the data used to represent a pixel of POL-SAR image are the coherency matrix itself without any manual feature extraction. It is obvious that the diagonal elements of coherency matrix are real values, while the nondiagonal elements are the complex numbers. Hence, the real and imaginary parts of the coherency matrix are picked out to constitute a vector. Considering the conjugate symmetry of coherency matrix, a 9-D column vector can be used to represent a pixel in the POL-SAR image

$$I = \left[2A_0, B_0 + B, B_0 - B, C, H, E, D, G, F\right]^T \quad (4)$$

where T denotes the transpose operator. Huynen [39] revealed that the elements in the vector have some physical meanings. Detailed information can be referred to [39]. The ith pixel is denoted by I_i. Symbol Θ denotes the set of POL-SAR image pixels, i.e., $\Theta = \{I_1, I_2, \ldots, I_N\}$, where N is the total number of POL-SAR pixels. Through the above-described steps, the complex coherence matrix of each pixel in the POL-SAR image is transformed into a 9-D real column vector. Finally, all the pixels in the image are reassembled into a POL-SAR data matrix. Thus, the data here used are a $9 \times N$ matrix.

Then, the distribution of POL-SAR data matrix is analyzed as follows. The first three elements of the vector are corresponding to the diagonal elements of coherency matrix. Therefore, the data distribution of them can be obtained by studying that of the diagonal elements in the coherency matrix. In line with a reciprocal theorem, the elements in the scattering matrix can be rearranged as a complex vector $u = [\,S_{hh}, S_{hv}, S_{vv}\,]^T$, which follows a multivariate complex Gaussian distribution [4]. Elements in u are expressed by complex numbers: $u_j = x_j + iy_j$, among which $j \in \{1, 2, 3\}$. The real and imaginary parts of any two complex elements in u are assumed to follow a circular Gaussian distribution [40], which requires that x_j and y_j for $j = 1, 2, 3$ have joint Gaussian distribution. It is easily proved in [33] that the real and imaginary parts of $e = (1/\sqrt{2})(S_{hh} + S_{vv}) = (1/\sqrt{2})[(x_1 + x_3) + i(y_1 + y_3)]$ independently follow the identical Gaussian distribution $N(0, \sigma^2)$. Consequently, the real and imaginary parts of $(1/\sigma)e$ follow the identical standard Gaussian distribution $N(0, 1)$ independently. Then, $|(1/\sigma)e|^2$ is regarded as subordinate to a χ^2-distribution of freedom 2. This distribution commonly used in mathematical statistics is also the Gamma distribution $\Gamma(1, 1/2)$. Thus, $|e|^2$ follows the Gamma distribution $\Gamma(1, \sigma^2/2)$. In the same way, it can be deduced that $|f|^2$ and $|g|^2$ obey the Gamma distribution. For the multilook POL-SAR data, $T = (1/L)\sum_{l=1}^{L} T(l)$, where $T(l)$ is the lth look and L is the total number of looks. These $T(l), l = 1, 2, 3, \ldots, L$ follow the independent and identically distribution. Hence, the first three rows of POL-SAR data matrix follow the Gamma distribution with shape parameter L. As a typical representative, histogram of the first row of POL-SAR data matrix is shown in Fig. 1(a), which verifies the correctness of the earlier analysis. The POL-SAR data set used here is the Flevoland image, whose details will be discussed in Section IV.



(a) (b) (c)

Fig. 1. Histograms of the three representative variables, i.e., $2A_0$, H, and F on the Flevoland image.

As we all know, if multiple independent random variables obey the standard normal distribution, the new random variable, which constituted by the sum of squares of these variables, follows the chi-square distribution. However, the product of two normally distributed variables does not follow chi-square distribution or Gamma distribution [41]. Actually, nondiagonal elements of coherency matrix are complex numbers. It is difficult to distinguish which kind of distribution the complex element obeys. What is really happening is that the nondiagonal elements of coherency matrix have been divided into the real and imaginary parts that are used to rearrange as the last six rows of POL-SAR data matrix. Therefore, it is impossible to analyze their distributions by utilizing the explanation method of data distribution of the first three rows of POL-SAR data matrix. We should consider their data distributions from another point of view. It is well known that histogram is an important and commonly used statistical method to analyze the distribution of data. The shape of histogram of all the last six rows of POL-SAR data matrix shows a symmetrical structure with zero as the center. This is very similar to the shape of the histogram of Gaussian distribution. Hence, it can be considered that the last six rows of POL-SAR data matrix are roughly subjected to the Gaussian distribution. In addition, as shown in [33], the Gaussian distribution is directly used to fit the remaining six rows of the POL-SAR data matrix. Thus, one of the real parts and one of the imaginary parts are selected as two representatives to display their histograms due to the limitation of space. The histogram of H and F is shown in Fig. 1(b) and (c), respectively. Through the analysis of the data distribution mentioned earlier, POL-SAR data matrix is considered to obey a mixture distribution of Gamma and Gaussian.

At length, the normalization method is used to preprocess the data matrix. According to the above-mentioned steps, we can adopt the reverse thinking process to normalize the POL-SAR data matrix in order to complete the process of data pretreatment. The first three rows of POL-SAR data matrix are divided by the corresponding variance in order to make the first three rows satisfying the chi-square distribution, while the last six rows are divided by the standard deviation so that the last six rows of POL-SAR data matrix follow the standard normal distribution. Then, the first three lines of the normalized data matrix are obedience to the chi-square distribution, which is also a special case of Wishart distribution in $p = 1$. Therefore, the first three rows of the POL-SAR data are regarded as a single variable Wishart distribution. From the point of view of the composition of multilook coherency matrix, they can also be looked as subjected to a univariable Wishart distribution.

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The last six rows are subjected to the standardized normal distribution. This normalization method is very important for the following work. In the following works, the POL-SAR data matrix is processed by this normalization method first.

III. Proposed Methods

A. Proposed MAE Method

The AE [10] is an unsupervised NN to learn features from a data set. It consists of two processes of encoder and decoder. The encoder is a function that maps the input signal $x_i \in \Re^n$ to the hidden representation $h_i \in \Re^m$, where n is the dimension of the input signal, m is the number of hidden layer nodes, and i is the index of the ith data. It has the following form:

$$h_i = f(W^{(1)}x_i + b^{(1)}) \tag{5}$$

where f is a nonlinear activation function, $W^{(1)} \in \Re^{m\times n}$ is the encoding weight matrix, and $b^{(1)} \in \Re^m$ is the encoding bias. We choose a logistic function: $f(z) = (1 + e^{-z})^{-1}$. Similarly, the decoder is a function that maps the hidden representation h_i to the reconstruct y_i

$$y_i = g(W^{(2)}h_i + b^{(2)}) \tag{6}$$

where g is also a nonlinear activation function, which is the same as that of the encoder process. $W^{(2)} \in \Re^{n\times m}$ is the decoding weight matrix, $b^{(2)} \in \Re^n$ is the decoding bias, and n represents the number of the output layer nodes. AE aims to get features by minimizing the reconstruction error of the likelihood function

$$L(X, Y) = \frac{1}{2}\sum_{i=1}^{N} \|y_i - x_i\|_2^2 \tag{7}$$

where X is the data matrix of training samples, Y is the reconstructed data matrix, and N indicates the number of training samples.

The number of hidden layer nodes should be greater than the input dimension in order to get an overcomplete feature set. Therefore, a sparse AE is selected to learn the features of the POL-SAR data matrix. The KL divergence is introduced as a penalty term. It can be written as

$$KL\left(\rho\|\hat{\rho}_q\right) = \rho\log\frac{\rho}{\hat{\rho}_q} + (1-\rho)\log\frac{1-\rho}{1-\hat{\rho}_q} \tag{8}$$

where ρ is the sparsity parameter and $\hat{\rho}_q$ means the average activation of the qth hidden node. By imposing sparsity on the hidden nodes during the training process, an AE network can learn useful structures in the input data, which has been verified through different references [42], [43].

The difference between the AE and WAE model mainly lies in the construction of data error term. The AE uses Euler distance for all pixels in the POL-SAR data matrix to measure errors. As a matter of fact, the coherency matrix of POL-SAR obeys the complex Wishart distribution. Hence, the Wishart distance is used to estimate the data error in the WAE [31]. The input and output of the process of the AE are all vectors. Before computing the gradient of the error function with respect to the parameters, the data form needs to be converted into the complex matrix. Therefore, the WAE involves the transformation of data forms between vector and complex matrix. According to different distribution characteristics of POL-SAR data matrix, a more reasonable loss function is constituted in the processing of AE. After POL-SAR data matrix preprocessed by the method aforementioned, the distribution characteristics of the data matrix satisfies a mixture distribution of Gaussian and Wishart. Therefore, we can utilize the Frobenius norm to the last six dimensions of the data matrix while taking advantage of the Wishart distance on the diagonal matrix, which is made up of the first three dimensions of the data matrix. As a matter of fact, the first three dimensions of data can be considered as subject to a single variable Wishart distribution. In this way, it can be further simplified to calculate the error of their respective Wishart distances, which eliminated the transformation process of data forms in the WAE model and improved the efficiency of calculation. To sum up, the proposed MAE is meant to roughly model the distribution of POL-SAR data matrix as the mixture distribution in a visible layer during the processing of AE.

Thus, the loss function of the proposed MAE consists of four parts: two types of data error term, a weight decay term, and a sparsity penalty term. It can be represented as

$$\begin{aligned} L(X, Y) = {} & \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{3} d\left(x_{ij}, y_{ij}\right) + \frac{1}{2N}\sum_{i=1}^{N}\sum_{j=4}^{9}\|y_{ij} - x_{ij}\|_2^2 \\ & + \frac{\alpha}{2}\sum_{l=1}^{2}\sum_{p=1}^{n}\sum_{q=1}^{m}\left\|W_{pq}^{(l)}\right\|_F^2 + \beta\sum_{q=1}^{m} KL(\rho\|\hat{\rho}_q) \end{aligned} \tag{9}$$

where $d_w(x_{ij}, y_{ij}) = Tr(x_{ij}^{-1}y_{ij} + \ln|x_{ij}|)$, α is the weight of weight decay term, β is the weight of sparsity penalty term, N is the number of input data, and $\|\bullet\|_F$ means the Frobenius norm.

The construction of two data error terms in Formula (9) is on the basis of the distribution of POL-SAR data matrix. To be specific, the first data error term is measured by Wishart distance and designed for the first three rows of data matrix, which obeyed the Wishart distribution, and the second data error term is calculated with Euclidean distance for the remaining rows of the data matrix, which followed the Gaussian distribution. Weight decay term is added in order to lessen the update speed of connected weights and prevent overfitting. The aim of introducing the sparsity penalty term is to make better use of the characteristics of the input data and the structural information of the network. The last two parts are consistent with the construction of loss function of sparse AE network.

B. How to Optimize the MAE

Just the same as the standard networks, the BP algorithm is applied to compute the gradient of the error function with respect to parameters. We need to initialize each parameter randomly and apply the optimization algorithm on the training data to achieve the goal of minimizing the loss function. From the above-mentioned loss function, it can be seen that the only difficulty is to solve the first term. Actually, it is easy

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to manage the remaining three items referring to the standard AE algorithm. In other words, it is important to optimize the first term as a function of weight and bias.

First, the decoding weight and bias are updated by the stochastic gradient descent algorithm

$$W^{(2)} = W^{(2)} - \eta \frac{\partial L(X,Y)}{\partial W^{(2)}} \tag{10}$$

$$b^{(2)} = b^{(2)} - \eta \frac{\partial L(X,Y)}{\partial b^{(2)}} \tag{11}$$

where η is the learning rate.

Then, the key step is to calculate the partial derivatives of the first term in the loss function of the proposed MAE. Hence, they can be expressed as

$$\frac{\partial d_{\text{wis}}(x_{ij}, y_{ij})}{\partial W^{(2)}} = \frac{\partial d_{\text{wis}}(x_{ij}, y_{ij})}{\partial y_{ij}} \frac{\partial y_{ij}}{\partial W^{(2)}} \tag{12}$$

$$\frac{\partial d_{\text{wis}}(x_{ij}, y_{ij})}{\partial b^{(2)}} = \frac{\partial d_{\text{wis}}(x_{ij}, y_{ij})}{\partial y_{ij}} \frac{\partial y_{ij}}{\partial b^{(2)}}. \tag{13}$$

With the function of Wishart distance, the partial derivatives mentioned earlier can be computed with the following equations:

$$\frac{\partial d_{\text{wis}}(x_{ij}, y_{ij})}{\partial y_{ij}} = \frac{\partial Tr(x_{ij}^{-1} y_{ij} + \ln |x_{ij}|)}{\partial y_{ij}} = x_{ij}^{-1} \tag{14}$$

$$\frac{\partial y}{\partial W^{(2)}} = \frac{\partial g(W^{(2)}h + b^{(2)})}{\partial W^{(2)}} = g'(W^{(2)}h + b^{(2)})h \tag{15}$$

$$\frac{\partial y}{\partial b^{(2)}} = \frac{\partial g(W^{(2)}h + b^{(2)})}{\partial b^{(2)}} = g'(W^{(2)}h + b^{(2)}). \tag{16}$$

Through substituting formulas (14) and (15) into formula (12), the partial derivative of Wishart distance on decoding weight can be obtained. Similarly, the partial derivative of Wishart distance on decoding bias can also be represented by substituting formulas (14) and (16) into formula (13). Combined with the last three terms, partial derivatives of the whole function are carried out by the following equations:

$$\frac{\partial L(X,Y)}{\partial W^{(2)}} = \delta^2 h^T / N + \alpha W^{(2)} \tag{17}$$

$$\frac{\partial L(X,Y)}{\partial b^{(2)}} = \delta^2 / N \tag{18}$$

$$\delta^2 = \left(\sum_{i=1}^{N} \sum_{j=1}^{3} x_{ij}^{-1} + \sum_{i=1}^{N} \sum_{j=4}^{9} (y_{ij} - x_{ij}) \right) g' \times (W^{(2)}h + b^{(2)}). \tag{19}$$

Then, formulas (10) and (11) will be updated with their partial derivatives.

Finally, the update of encoding weight and bias is similar to those of the decoding process. Thus, we will not illustrate them in detail. The result is directly given as follows:

$$W^{(1)} = W^{(1)} - \eta(\delta^1 x^T / N + \alpha W^{(1)}) \tag{20}$$

$$b^{(1)} = b^{(1)} - \eta(\delta^1 / N) \tag{21}$$

$$\delta^1 = \left(W^{(2)}\delta^2 + \beta \left(-\frac{\rho}{\hat{\rho}_q} + \frac{1-\rho}{1-\hat{\rho}_q} \right) \right) f'(W^{(1)}x + b^{(1)}). \tag{22}$$

TABLE I

CLASSIFICATION OF POL-SAR IMAGE BASED ON THE PROPOSED MAE

Algorithm1: Classifier Based on MAE
Input: The POL-SAR data matrix and its corresponding ground-truth; The numbers of hidden nodes and categories; Sparsity parameter; Weights of weight decay term and sparsity penalty term;Parameter of weight decay term of Softmax classifier. 1: Preprocess all POL-SAR pixels by being divided the corresponding standard deviation of each dimension; 2: Randomly selected 1% data as the unlabeled data and 5% labeled data as the training data; 3: With unlabeled data, the MAE is trained according to (14)-(24); 4: Get the parameters corresponding to the encoding process in MAE and learn the features of training data; 5: Train the Softmax classifier with the learned features and the corresponding label information; 6: Classify the whole POL-SAR image pixel by pixel with the trained Softmax classifier. Output: Classification result.

C. Combination of the MAE With a Classifier

After obtaining the parameter of the MAE, they could be utilized to extract the features of POL-SAR image. Since the information of interest is contained within the activation of hidden units, the value of hidden nodes is employed as the features of an image. In order to complete the classification task of POL-SAR image, a small amount of labeled pixels are selected as training samples for the classifier. Specifically, features obtained by the trained MAE and the corresponding label information are used to train a classifier. In our experiments, a softmax classifier [44] trained by the L-BFGS [45] algorithm is chosen and connected with the proposed MAE.

The softmax classifier is a generalization of the polynomial logistic function. It generates a K-dimensional real-valued vector in the range of 0–1, which represents a categorical probability distribution. Function (23) shows how softmax function predicts the probability for the class given a sample vector X

$$h_w(X) = P(y = j|X; W) = \frac{e^{X^T W_j}}{\sum_{k=1}^{K} e^{X^T W_k}} \tag{23}$$

where K is the total number of categories, W_k represents the weight vector of the class k, and W is the weight matrix of the softmax classifier.

In a word, features of each pixel in POL-SAR image, which extracted by the proposed MAE, are input into the trained softmax classifier to obtain the classification results. As shown in Table I, the classification of POL-SAR image based on the proposed MAE is summarized in Algorithm 1.

D. Classification Network Based on the Proposed SMAE

The SAE constructs its network framework with multiple sparse autoencoders, in which the output of each layer is wired to the inputs of the successive layer. In addition, the SAE is trained layer by layer. It can be regarded as a process of features' extraction from low level to high level. From the point of view of features' learning, the information of deeper layer is more beneficial to the classification task. This multiple-layer modeling method leaves out the process of

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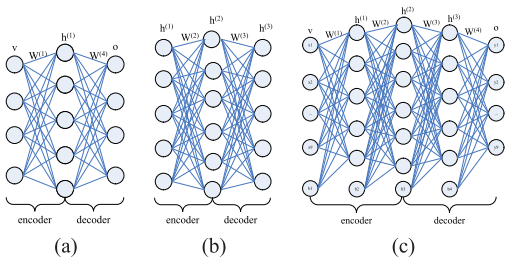

Fig. 2. (a) and (b) Structure of the AE. (c) Structure of the SAE combined by the former two AEs.

manual feature extraction and can be regarded as a process of multiple-layer learning representation. It is shown that the learned high-level representations capture a lot of structures in the input data (such as the MNIST and the Iris flower database) [23], which are very useful for specific tasks, e.g., classification and information retrieval. For the sake of simplicity, a particular SAE network stacked by two AEs is shown in Fig. 2 with one visible layer, three hidden layers, and an output layer.

Let us suppose that the proposed MAE is used to build an SAE network. The visible layer of the second MAE is the hidden layer of the previous MAE, the output of which follows the Bernoulli distribution [33]. At this point, the data distribution of input is no longer consistent with the requirement of the MAE. Therefore, it may be unsuitable to connect two MAEs to compose the SAE network. In our scheme, the proposed MAE is used as the first AE, whereas the second one is a standard AE. This method may be an appropriate choice for feature learning of POL-SAR image. The AE shown in Fig. 2(a) with visible layer v and hidden layer $h^{(1)}$ is the proposed MAE, where v and $h^{(1)}$ follow the mixture distribution and Bernoulli distribution, respectively, while the AE shown in Fig. 2(b) is an ordinary AE with visible layer $h^{(1)}$ and hidden layer $h^{(2)}$, in which both $h^{(1)}$ and $h^{(2)}$ follow the Bernoulli distribution. As a result, these two layers in different AEs follow the same distribution and can be simply combined to form an SAE, which named SMAE in this article. Up to now, the SMAE NN is set up, as shown in Fig. 2(c). It can also be trained in an entirely unsupervised way, which makes the best of the unlabeled POL-SAR pixels. From the point of view of feature learning, it maps the original input vector into a two-layer feature space.

The features of POL-SAR pixels learned by SMAE can be used as the input for many classification methods. Softmax classifier is selected in order to adjust the learned features easily. For a simple explanation, we only choose two AEs to build the SAE network. As shown in Fig. 3, the schematic of a four-layer classification network is presented here, which consists of an input layer, two hidden layers, and an output layer, and the corresponding flowchart is also given. The first three layers are corresponding to the encoding process of SMAE, and the last output layer with sigmoid function plays a role of softmax classifier. First, we can use the unlabeled data to train the SMAE and obtain the initial parameters. Then, we can also achieve the initial parameters of the softmax

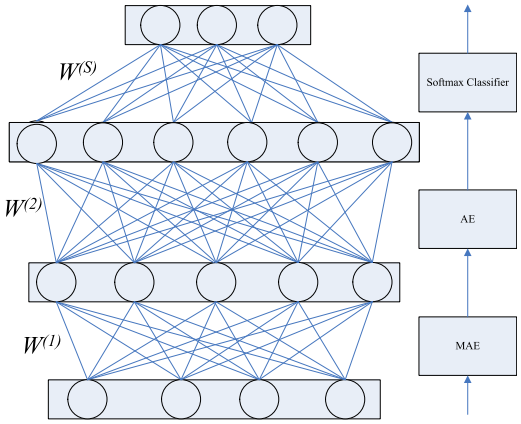

Fig. 3. Classification network initialized by the weights of SMAE and the corresponding flowchart.

TABLE II
CLASSIFICATION NETWORK BASED ON SMAE

Algorithm2: Classification network based on SMAE

Input: The POL-SAR pixel set and its corresponding ground-truth; The numbers of hidden unitsand categories; Sparsity parameter and ; Weights of weight decay term and sparsity penalty term ; Parameter of weight decay term of Softmax classifier.
1: Preprocess all pixels by being divided the corresponding standard deviation of each dimension;
2: Randomly selected 1% data for unsupervised training of SMAE and 5% labeled data as the training data;
3: With unlabeled pixel set, the first MAE is trained according to (14)-(24), leading to the parameters ;
4: Put pixels into the visible units of the trained MAE and sample data from the hidden units; train the second AE with these sampled data; the parameters of the second AE are obtained ;
5: Get the parameters corresponding to the first three layers in SMAE;
6: Learn features using the parameters in the above step and train the Softmax classifier with label information, getting the parameter ;
7: Initialize parameters of the proposed classification network and train the network by BP algorithm with the training data;
8: Classify the whole POL-SAR image pixel by pixel.
Output: Classification result.

classifier by supervised training with the label information of a small number of labeled POL-SAR pixels and the features extracted by the SMAE network with discarding the decoding process. Finally, with the parameters of the trained SMAE and the parameters of the softmax classifier as the initial parameters of the proposed classification network, a BP algorithm is implemented to fine-tune the parameters so that the classification network can be adjusted to fit the task well. In short, a classification network based on the SMAE is summarized in Algorithm 2, as shown in Table II.

IV. EXPERIMENTAL ANALYSIS AND RESULTS

A. Comparison Methods

In the following experiments, a classifier based on the proposed MAE is compared with three classifiers, including Wishart classifier and classifiers based on AE network and WAE model. In the same way, the SVM model and two classification networks with a similar classification process are selected as three comparison methods of the classification network based on the proposed SMAE.

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Wishart classifier classifies each pixel according to the distance between the pixel and each center. The performance of the algorithm is determined by the quality of the center [2]. The AE network uses the F-norm to measure the similarity of the input and the output [31]. This similarity measurement method is suited for natural images that obey the Gaussian distribution. The WAE model [31] takes the distribution of original data into account using the AE network to extract the features of POL-SAR data. Hence, it is especially suitable for POL-SAR data that follow complex Wishart distribution. Both the models can be regarded as a process of feature learning. It is well known that the performance of image classification is heavily dependent on its features. Therefore, a classifier connected with the feature extraction model is designed to execute the classification task. Because of the similar procedure for classification task, the aforementioned classifiers are selected as comparison methods for classifier based on the proposed MAE.

SVM is used as a comparison method for its good classification performance [16]. In [46], an effective network method for POL-SAR data classification was proposed, in which the SAE is applied as a useful strategy to achieve the feature learning and then connected with a softmax classifier to form a classification network. On the one hand, classification network based on the SAE has something in common with classification network based on the proposed SMAE, such as the composition of input data, network structure, and the method of training. On the other hand, there are several differences, including the way of data preprocessing and the selection of training data. What is more, the SAE does not consider the distribution characteristics of POL-SAR data matrix and the consistency of data distribution between the output of the former layer and the input of the next hidden layer. Therefore, the SAE directly cascades multiple AEs to fulfill the process of feature extraction. For a fair comparison, we also give the classification network based on the stacked WAE (SWAE), in which all the manipulations are consistent with those of classification network based on the proposed SMAE, except the form of input data. It mainly considers the Wishart distribution of the coherent matrix of POL-SAR, and the key point is to deal with the transformation of data format in different stages of the optimization process. In brief, the advantages of the SMAE are fully considering the distribution of data at different stages and exploiting the unlabeled samples at utmost to accomplish the training process of feature learning.

B. Training Data Ratio

In the experiments of the supervised Wishart classifier and SVM model, 5% labeled data are chosen as the training data. Nevertheless, the following three classifiers, which based on the feature extraction model, have a few differences in the selection of training data. Due to the training of AE, WAE and the proposed MAE are unsupervised feature learning process, and 1% pixels are selected as the training set, regardless of whether the data are labeled or not. After that, 5% labeled data are used to train the softmax classifier. This is done in accordance with the comparison methods in the literature [31]. In our experiments, we randomly select 5% rather than 10% labeled data as the training samples for the implementation of classification network based on SAE, which is the only difference with the literature [46]. However, different from the abovementioned method, we choose 1% pixels to train the proposed SMAE feature learning. Then, 5% labeled data with the label information are used to train the softmax classifier. With parameters obtained by the above-mentioned processes as initial parameters, the labeled data are used to train the classification network based on SMAE at last. In our experiment, the setting method of training data of classification network based on SWAE is consistent with that of classification network based on the proposed same, because both of them have the same procedure for accomplishing the task of classification. In addition, instead of the residual 95%, all the labeled data are chosen as the testing data in all the experiments for the convenience of calculating the evaluation indexes and displaying the classification result image corresponding to the ground-truth map.

C. Parameter Selection

Three real POL-SAR images are used to verify the performance of the proposed methods, including images for Flevoland, The Netherlands; Oberpfaffenhofen, Germany; and San Francisco Bay, CA, USA. For Wishart classifier, the average of the selected training samples corresponding to each class is set as the center of each class, and the Wishart distance is used to classify every pixels. Therefore, none of the parameters need to be set. In order to compare with the AE and WAE model and illustrate the effectiveness of the proposed MAE, we mainly discuss the selection of parameters of three classifiers based on feature extraction. These classifiers consist of five parameters. According to the discussion of the parameters in other articles [31], [46], we set the default value of the weight of sparsity penalty term and weight decay term as 1 and 1e−6, respectively. In the following experiments, we will focus mainly on the selection of the number of hidden layer nodes, sparsity parameter, and the parameter of softmax classifier.

At first, the influence of the selection of the softmax classifier parameter value on the effect of the three classifiers is discussed. To do that, other two parameters are fixed the same as in [31], in which the number of hidden layer nodes in the AE stage is 100 and sparsity parameter is 0.05. We also make use of the same network structure and parameters' setting for the AE, WAE, and MAE in each experiment. The overall accuracy (OA) corresponding to the softmax classifier parameter values on the three classifiers over three POL-SAR images is shown in Fig. 4. OA refers to the ratio of the number of pixels being classified correctly to the total number of labeled pixels in the ground-truth map. It is known from Fig. 4(a)–(c) that the OA curve increases with increase in the −Log(Lambda) value and tends to be stable. This shows the effectiveness of the three classifiers. In addition, the OA curves of classifiers based on WAE and the proposed MAE are smooth, which is on behalf of the better robust to the

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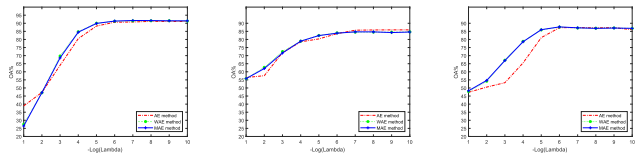

(a) (b) (c)

Fig. 4. OA with parameters values of softmax classifier over the three POL-SAR images. (a) Flevoland, (b) Oberpfaffenhofen, and (c) San Francisco Bay given by the three methods.

selection of the parameter. However, compared with the other two classifiers in Fig. 4, there is a small fluctuation in the OA curve of classifier based on AE, which indicates that the classifier based on AE could be easily influenced by the selection of parameter values. To sum up, values of Lambda below 1e−5 for the three classifiers always have a good effect on the three POL-SAR test images. Therefore, 1e−7 is selected as the default value of Lambda to carry out the three classifiers in the following experiments.

Given the parameter of softmax classifier, three classifiers adopt the way of grid search on the number of hidden nodes and sparsity parameter over the three test images and the OA on the combination of different parameters is also provided. Table III shows the classification OA of different methods under different pairs of parameters over the Flevoland image, in which the highest value of each pair of parameters is rendered with bold. It is obvious that the proposed MAE has a good effect on most pairs of parameters. For different combinations of parameters, the OA of classifier based on the AE network is changing between 83% and 91%, while the OA of classifiers based on the WAE and MAE network varies with small amplitude at 91% and 92%, respectively. In other words, the AE is less robust to the selection of the number of hidden nodes and sparsity parameter than the others. Furthermore, it is known from the third column that compared with other classifiers, the OA of classifier based on AE has a larger fluctuation with the change of sparsity parameter, given the number of nodes in the hidden layer. When the number of hidden layer node reaches 300, there are too many parameters to be trained in the AE network, which led to a lower OA than that of other network structures. From the comparison between the last row and other rows, it can be seen that the OA obtained by the softmax classifier utilizing all features is often lower than those acquired by softmax classifier using part of features in most cases. This shows the necessity of introducing sparse penalty term into the loss function. Therefore, sparsity parameter with 1.00 and the number of hidden nodes with 30 and 300 are not the appropriate parameters for three classifiers simultaneously in the experiments. At last, any pair of parameters in the remaining could be selected as the default value for further experiments.

From Table IV, it is clear that the three classifiers have a similar performance on each pair of parameters for Oberpfaffenhofen image. However, according to the third and last columns of Table IV, the OA of classifier based on AE is a bit lower than that of the other classifiers. Similar to the previous image, the WAE model and the proposed MAE are more robust than the AE network on the selection of sparsity parameter and the number of hidden nodes in the network structure. The OA of all three classifiers is higher for some pairs of parameters, which is shown with bold in Table IV. Therefore, we can choose a suitable pair of parameters to carry out the follow-up studies.

Table V gives the result of three classifiers on different pairs of parameter over the San Francisco Bay image. Similar to the two experiments mentioned earlier, three classifiers may always have a good classification performance on the number of hidden nodes lower than 300 and the sparsity parameter higher than 0.01. As a result, we are no longer making a detailed analysis.

For the SVM model, grid search is used to obtain the kernel parameters of radial basis function. Three classification networks based on SAE, SWAE, and SMAE in our experiment use the same network structure as in [46]. Concretely, 30 and 60 are chosen as the dimension of the corresponding two hidden layers, respectively. Due to a considerable number of parameters contained in these classification networks, we will

TABLE III

OAs (%) OF THREE CLASSIFIERS WITH DIFFERENT VALUES OF HIDDEN SIZE (N) AND SPARSITY (ρ) OVER THE FLEVOLAND IMAGE

Methods	OA% N / ρ	30	80	100	150	300
AE	0.01	87.65	87.93	89.44	90.36	88.33
WAE		91.46	91.43	91.77	91.28	91.61
MAE		**91.96**	**92.05**	**91.97**	**91.77**	**92.07**
AE	0.05	84.41	90.94	91.38	89.80	87.10
WAE		91.57	91.67	91.55	91.67	91.60
MAE		**92.03**	**92.04**	**92.04**	**92.16**	**92.04**
AE	0.10	84.32	91.09	91.34	90.51	88.02
WAE		91.47	91.63	91.59	91.59	91.58
MAE		**92.07**	**92.13**	**92.04**	**92.12**	**92.05**
AE	0.50	87.14	91.27	91.36	90.12	87.41
WAE		91.40	91.35	91.33	91.31	91.45
MAE		**92.04**	**91.89**	**91.55**	**91.59**	**92.07**
AE	1.00	83.00	**91.25**	**91.14**	89.11	86.11
WAE		89.78	89.81	89.63	89.82	**90.04**
MAE		**89.87**	89.75	89.60	**89.89**	89.29

TABLE IV

OAs (%) OF THREE CLASSIFIERS WITH DIFFERENT VALUES OF HIDDEN SIZE (N) AND SPARSITY (ρ) OVER THE OBERPFAFFENHOFEN IMAGE

Methods	OA% N / ρ	30	**80**	**100**	**150**	300
AE	**0.01**	82.13	84.84	86.05	84.83	83.89
WAE		84.43	84.37	84.47	84.66	84.53
MAE		84.70	84.52	84.54	84.77	84.66
AE	**0.05**	82.03	85.24	85.40	84.99	84.03
WAE		84.41	84.38	84.56	84.52	84.51
MAE		84.65	84.62	84.57	84.65	84.61
AE	**0.10**	82.26	85.24	85.75	84.59	84.06
WAE		84.42	84.22	84.26	84.51	84.37
MAE		84.59	84.58	84.53	84.58	84.62
AE	**0.50**	82.38	85.09	85.95	84.69	83.62
WAE		84.12	84.34	84.32	84.35	84.38
MAE		84.31	84.34	84.47	84.32	84.38
AE	1.00	81.32	83.26	84.87	84.35	83.75
WAE		83.75	83.89	83.61	83.73	84.02
MAE		84.01	83.60	84.02	83.59	83.93

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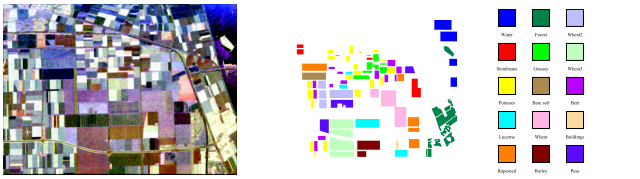
TABLE V

OAs (%) OF THREE CLASSIFIERS WITH DIFFERENT VALUES OF HIDDEN SIZE (N) AND SPARSITY (ρ) OVER THE SAN FRANCISCO BAY IMAGE

Methods	OA% \ N / ρ	30	**80**	**100**	**150**	300
AE		81.32	86.92	87.21	88.58	86.08
WAE	**0.01**	88.13	87.97	87.99	88.19	88.05
MAE		88.00	88.34	88.20	87.97	88.22
AE		68.97	87.71	88.32	88.13	86.46
WAE	**0.05**	87.80	87.55	87.76	87.69	87.50
MAE		87.55	87.98	87.72	87.19	87.65
AE		78.10	87.04	88.34	88.04	86.09
WAE	**0.10**	87.27	87.20	87.31	87.35	87.14
MAE		87.53	87.47	87.65	87.16	87.80
AE		68.33	86.04	88.42	88.01	86.37
WAE	**0.50**	87.04	87.07	86.85	86.79	86.76
MAE		86.61	86.85	86.86	86.90	86.73
AE		56.75	85.17	88.25	86.29	86.10
WAE	1.00	86.25	83.60	83.58	86.57	86.33
MAE		86.02	86.63	86.27	86.65	86.22



(a) (b) (c)

Fig. 5. (a) Pauli RGB image of dataset Flevoland. (b) Corresponding ground-truth map. (c) Terrains.

no more make a discussion on the selection of parameters in detail here. Because the classification network is made up of different functional components, we could set up parameters of the four-layer network according to the experience of parameter selection mentioned earlier or we can also refer to the parameter setting in other related classification methods. In the following experiments, details of parameters' setting for different methods are given over three test images.

D. Flevoland Image

The Flevoland image is obtained from a subset of an L-band, multilook POL-SAR data, acquired by the AIR-SAR platform on August 16, 1989, with a size of 750 × 1024. The Pauli RGB image and the corresponding ground-truth map are shown in Fig. 5. The total number of labeled pixels in the ground truth is 167 712. There are 15classes in this ground truth, in which each class indicates a type of land covering and is identified by one color.

Wishart classifier is a method of supervised classification. The only thing to be done is to give the training samples set and the number of iterations. For AE, WAE, and the proposed MAE, we select 100 as the number of hidden nodes to complete the process of feature learning. In this experiment, the sparsity parameter and the weight of sparsity penalty term are set as 0.05 and 1, respectively. Moreover, 1e−6 is used as the weight of weight decay term and the weight parameter of softmax classifier. SVM is also a supervised classification model. In the implementation process with LIBSVM, the RBF is chosen as the kernel function. The setting of gamma parameter in kernel function needs to be searched within a certain range in order to achieve a better classification result. For a fair comparison, the number of nodes within two hidden layers is set as 30 and 60 to construct SAE, SWAE, and the proposed SMAE, which is the same as in [46]. As described in [46], sparsity parameter of the second hidden layer tends to be lower than that of the first, which leads to a better classification effect. Therefore, we use the same parameter setting of SAE as in [46] in our experiments. Furthermore, the remaining parameters are consistent with those of the proposed SMAE. For the SWAE and the proposed SMAE, we choose the same sparsity parameter for two hidden layers, which is different from the SAE. In this experiment, 0.05 and 1e−6 are selected as sparsity parameter and the weight of softmax classifier, respectively. Besides, we also use 1e−6 and 1 as the weight of the weight penalty term and the weight of the sparsity penalty term, respectively. Then, we analyze the classification result of the proposed classification methods and make a comparison with the corresponding compared methods.

First, we talk over the classification result on the Flevoland image of the classifiers based on Wishart distance, AE, WAE, and the proposed MAE. The performance of the compared classification methods can be evaluated via the objective quantitative indexes of the correct rate of each category, the OA, the training time (T_{training}) of the classification methods, and the predicting time ($T_{\text{predicting}}$) for all the pixels in the POL-SAR image and via the subjective qualitative analysis of visual results of the corresponding ground-truth image. Since the selection of training data is random, we choose to perform all the experiments 30 times here to better illustrate the performance of the comparison of classification methods. Furthermore, we adopt the strategy of discarding the data of the first ten maxima and the last ten minima to mitigate the impact of the random selecting of training data on the compared classification methods. As shown in Table VI, the performance of objective quantitative indexes is evaluated by both the means and the standard deviations of the remaining ten experiments, and the maximum value of each index for the four classifiers is expressed in bold. According to the data in Table VI, we can draw the following conclusions. First, the OA of Wishart classifier is the lowest in all methods. However, the correct rate of Bare soil is the highest. Most of all, the correct rate of Bare soil is 3% higher than the second highest value obtained by the classifier based on AE. Then, the following three classifiers, which used the same way to complete the task of classification, have a higher OA than Wishart classifier. Furthermore, the classifier based on the proposed MAE has a certain improvement in classification OA compared with other two classifiers. From the point of view of the correct rate for each land cover, the classifier based on the proposed MAE has the highest classification correct rate on most of them, while the correct rate of Water, Forest, Rapseed, and Wheat2 is a little lower than that of the other two classifiers. The OA of classifier based on AE is the lowest among the three classifiers based on feature learning. Finally, we analyze the implementation efficiency of the algorithm from the view of time. Wishart classifier directly

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TABLE VI
CLASSIFICATION PERFORMANCE OF THE CLASSIFIER WITH THE PROPOSED MAE AND ITS THREE COMPARING METHODS ON THE FLEVOLAND IMAGE

	Water	Stem beans	Potatoes	Forest	Grasses	Beet
Wishart	0.4978±0.0030	0.9263±0.0080	0.8690±0.0075	0.8743±0.0046	0.6843±0.0111	0.9155±0.0082
AE	**0.9869±0.0055**	0.9314±0.0108	0.8691±0.0133	0.8814±0.0107	0.8605±0.0097	0.9387±0.0085
WAE	0.9688±0.0077	0.9543±0.0082	0.8948±0.0061	**0.9118±0.0042**	0.8542±0.0096	0.9546±0.0064
MAE	0.9802±0.0096	**0.9591±0.0099**	**0.8964±0.0064**	0.9113±0.0050	**0.8641±0.0097**	**0.9577±0.0070**
	Rapeseed	Peas	Lucerne	Bare Soil	Wheat2	Wheat
Wishart	0.7231±0.0144	0.9404±0.0079	0.7871±0.0117	**0.9877±0.0011**	0.8183±0.0083	0.8624±0.0092
AE	**0.8486±0.0123**	0.9477±0.0112	0.9479±0.0075	0.9572±0.0043	**0.8224±0.0141**	0.9046±0.0103
WAE	0.8406±0.0113	0.9588±0.0050	0.9506±0.0042	0.9118±0.0127	0.8081±0.0077	0.9142±0.0068
MAE	0.8411±0.0115	**0.9642±0.0052**	**0.9534±0.0043**	0.9262±0.0168	0.8102±0.0078	**0.9157±0.0066**
	Wheat3	Buildings	Barley	OA	$T_{training}$ (s)	$T_{predicting}$ (s)
Wishart	0.8805±0.0035	0.8188±0.0061	0.9498±0.0063	0.8297±0.0005	0.00±0.00	**86.02±1.41**
AE	0.9498±0.0039	0.7627±0.0509	0.9548±0.0088	0.9113±0.0009	5.83±0.41	0.30±0.01
WAE	0.9506±0.0033	0.8409±0.0212	0.9579±0.0059	0.9161±0.0004	**18.08±0.56**	0.26±0.01
MAE	**0.9525±0.0030**	**0.8442±0.0249**	**0.9598±0.0062**	**0.9201±0.0003**	9.45±0.60	0.26±0.01

selects some labeled data as the initial classification center to finish the supervised classification. Thus, the training time is zero. In addition, the prediction time of Wishart classifier is quite long and it takes about 100 s. The time for training three classifiers based on feature learning is no more than dozens of seconds. In particular, the training time of classifier based on AE and WAE is about 5 and 18 s, which is the shortest and longest, respectively, while the time of classifier based on the proposed MAE is in the middle of the above-mentioned classifiers. However, the time consumed by predicting all the pixels using the classifiers based on feature extraction is less than 1 s. Besides, the prediction time is almost the same and closes to 0.3 s.

Next, the result image corresponding to the ground truth of Flevoland and its prediction image with all pixels, given by the classifier based on the proposed MAE and three comparing classifiers, are shown in Fig. 6(a)–(h. The images shown here are the result of one execution of all comparison methods. The black rectangles in Fig. 6(a) are used to highlight the remarkable misclassified pixels of the Wishart classifier, which represented for the Water, Grasses, and Wheat. Nevertheless, from the black ellipses in Fig. 6(a)–(d), which mainly used to indicate the Bare soil, it can be found that the classification result of the Wishart classifier is smoother than that of others. Classifier based on the proposed MAE has the best classification effect on Water, which marked with the black rectangles in Fig. 6(b)–(d). For the other ground objects, it is difficult to distinguish which result image is the best one between Fig. 6(c) and (d) with the naked eyes. However, it is obvious that both of them are better than the result image in Fig. 6(b). The predicted images shown in Fig. 6(e)–(h) validate the effectiveness of the proposed MAE and the rationality of the above-mentioned analysis a second time.

At length, the classification results of SVM and three classification networks based on SAE, SWAE, and the proposed SMAE are discussed. Table VII provides the evaluation indexes of each method in the same way. We can make conclusions from Table VII as follows. First, compared with classifiers based on feature extraction, the corresponding classification networks all have a certain improvement in classification OA. The OA of classification network based on the proposed SMAE is the highest among the four methods.

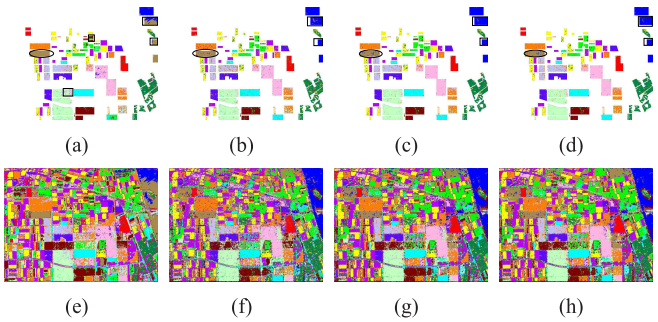

Fig. 6. Result image corresponding to the ground truth of Flevoland with (a) Wishart, (b) AE, (c) WAE, (d) MAE, and (e)–(h) their prediction image.

Besides, different methods have excellent classification effect on different ground objects. The correct rate of SVM on Barely is the highest. It also has a better classification effect on Water, Stem beans Beet, and Peas. However, the correct rate of SVM for the rest categories is worse than that of at least two of three classification networks based on different AE networks. Especially, the classification correct rate of Buildings is as low as about seven percentage points. Among all the methods, the classification network based on SWAE has the highest correct rate for Water, Bare soil, and Buildings. Besides, the maximum correct rate of them is no more 1% than those of the classification network based on the proposed SMAE. The classification network based on SAE is good at classifying the pixels of Potatoes and the OA of which is a bit higher than those of other two classification networks. The correct rate of classification network based on the proposed SMAE for the remaining 11 kinds of ground object is higher than that of classification network based on SWAE. With only three types of ground object, the correct rate of classification network based on SMAE is significantly higher than that of classification network based on SWAE. Without considering the process of parameter optimization, the training time of SVM is less than 95 s, while the time for training the classification networks based on different stacked AEs is a little longer, and it takes about 100 s, which is nearly ten times of the corresponding classifiers based on feature extraction. Specifically, the training time of classification network based

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TABLE VII

CLASSIFICATION PERFORMANCE OF THE CLASSIFICATION NETWORK BASED ON THE PROPOSED SMAE AND ITS THREE COMPARING METHODS ON THE FLEVOLAND IMAGE

	Water	Stem beans	Potatoes	Forest	Grasses	Beet
SVM	0.9710±0.0057	0.9492±0.0157	0.8251±0.0363	0.8862±0.0109	0.7847±0.0339	0.9519±0.0075
SAE	0.9852±0.0189	0.9265±0.0331	**0.8982±0.0086**	0.8980±0.0111	0.9068±0.0109	0.9545±0.0073
SWAE	**0.9951±0.0019**	0.9426±0.0096	0.8916±0.0090	0.8937±0.0092	0.9128±0.0094	0.9519±0.0063
SMAE	0.9943±0.0025	**0.9514±0.0080**	0.8949±0.0072	**0.8994±0.0077**	**0.9160±0.0095**	**0.9595±0.0059**
	Rapeseed	Peas	Lucerne	Bare Soil	Wheat2	Wheat
SVM	0.7737±0.0218	0.9610±0.0063	0.9515±0.0101	0.9077±0.0227	0.7892±0.0232	0.9125±0.0083
SAE	0.8645±0.0349	0.9494±0.0078	0.9591±0.0107	0.9685±0.0567	0.8251±0.0400	0.9236±0.0158
SWAE	0.8772±0.0084	0.9482±0.0065	0.9643±0.0046	**0.9896±0.0045**	0.8323±0.0106	0.9213±0.0065
SMAE	**0.8817±0.0077**	**0.9649±0.0050**	**0.9658±0.0041**	0.9887±0.0042	**0.8407±0.0097**	**0.9283±0.0068**
	Wheat3	Buildings	Barley	OA	$T_{training}$ (s)	$T_{predicting}$ (s)
SVM	0.9467±0.0049	0.0685±0.0189	**0.9705±0.0066**	0.8925±0.0016	93.31±9.89	**138.39±5.84**
SAE	0.9538±0.0078	0.8366±0.0331	0.9613±0.0103	0.9245±0.0005	80.09±11.50	2.61±0.06
SWAE	0.9533±0.0029	**0.8462±0.0269**	0.9656±0.0057	0.9263±0.0005	**106.49±1.64**	1.41±0.05
SMAE	**0.9538±0.0025**	0.8375±0.0258	0.9653±0.0073	**0.9308±0.0006**	93.65±2.07	1.40±0.04

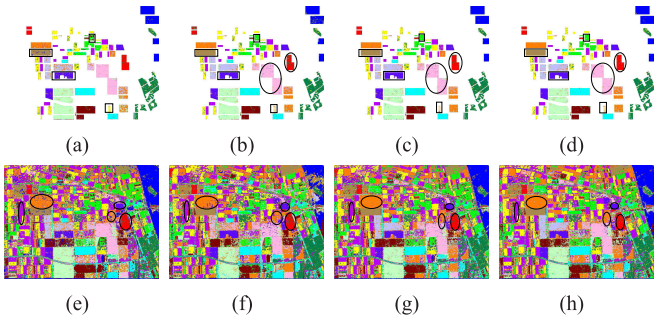

Fig. 7. Result image corresponding to the ground truth of Flevoland with (a) SVM, (b) SAE, (c) SWAE, (d) SMAE, and (e)–(h) their prediction image.

on SAE and SWAE is the shortest and longest and takes around 80 and 100 s, respectively. The time for accomplishing the training process of classification network based on the proposed SMAE is somewhere between the times spent on the abovementioned two classification networks. From the comparison of the variances, the stability of the latter two classification networks is better than that of the former. The time cost by predicting all pixels in the image with all three classification networks is almost the same and closes to a couple of seconds. However, the prediction time of SVM is relatively long and more than 100 s. Moreover, the prediction time needed for classification network based on the proposed SMAE is about 1.4 s, which is almost the same as that of the classification network based on SWAE and also the least among the three classification networks with the same structure.

Combined with the result images corresponding to the ground truth of Flevoland generated by four methods shown in Fig. 7(a)–(d), a detailed analysis of the classification effect on 15 kinds of ground objects is given. We emphasize the Bare soil, Wheat2, and Buildings with black rectangles in Fig. 7(a)–(d). From these blocks, it can be seen that the classification effect of SVM for these pixels is inferior to that of the three classification networks. In particular, the classification result of Buildings is hardly right, and almost all of them are divided into Forest. Among the three classification networks, classification network based on the proposed SMAE has the best classification performance on Wheat, Peas, and Stem beans, which circled by the black ellipses in Fig. 7(b)–(d). It is clear that the Water shown in the result image of Fig. 7(c) and (d) is smoother than that of Fig. 7(a) and (b). This illustrates that the classified ability to Water of classification networks based on SWAE and SMAE exceeds that of classification network based on SAE and SVM. However, the influence on Barley by four classification methods is just the opposite, according to the corresponding areas of classification result images. Furthermore, all the classification networks have a good classification effect on Beet, Lucerne, Wheat3, and Forest, while the classification performance of Potatoes, Rapeseed, and Wheat2 by four classification methods is not satisfactory. At last, the classification performance of SVM on Grasses is obviously worse than that of three classification networks.

Corresponding to SVM and three classification networks, Fig. 7(e)–(h) shows the prediction result of all the pixels in the Flevoland image. The prediction result of 15 categories in these images is fairly consistent with the abovementioned analysis on the results of the corresponding ground-truth images. On the whole, the boundaries between blocks are clear in the result images. However, the classification effect of boundary pixels is not as good as that of pixels in the block, and even the phenomenon of wrong classification often occurs. Sometimes, the classification result of boundary pixels is neither in this block nor in that. There are other classes of pixels in most of the blocks, which results in the noise and speckles of the image block. Generally, the smoother the image blocks, the higher the classification OA of crops. It can also be seen the importance of spatial neighborhood information is particularly prominent, which is very essential for solving the misclassified scatter pixels. From the blocks marked by the black oval shown in the predicted images in Fig. 7(e)–(h), we can find that the classification network based on the proposed SMAE has the best classification effect on Stem beans, Peas, Rapeseed, and Beet. For the great majority categories of the remaining, the classification performance of classification network based on the proposed SMAE is almost the same as that of classification network based on SWAE and

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TABLE VIII
CLASSIFICATION PERFORMANCE OF TWO PROPOSED METHODS AND SIX COMPARING METHODS FOR OBERPFAFFENHOFEN IMAGE

	Wood land	Open area	Built-up area	OA	$T_{training}$ (s)	$T_{predicting}$ (s)
Wishart	0.6550±0.0098	0.9514±0.0014	0.5952±0.0043	0.8056±0.0003	0.00±0.00	35.81±0.27
AE	0.7681±0.0070	0.9767±0.0012	0.6298±0.0075	0.8501±0.0005	5.22±0.23	1.37±0.03
WAE	0.7191±0.0072	0.9772±0.0007	0.6515±0.0044	0.8467±0.0004	35.22±0.56	1.35±0.04
MAE	0.7300±0.0111	0.9775±0.0008	0.6519±0.0047	0.8490±0.0005	14.38±1.07	1.34±0.03
SVM	0.7829±0.0398	**0.9792±0.0036**	0.6205±0.0333	0.8562±0.0023	65.37±9.85	**1173.48±130.69**
SAE	0.3724±0.3085	0.7315±0.2194	0.7456±0.1389	0.7005±0.0259	**526.18±121.60**	4.91±0.14
SWAE	0.8747±0.0026	0.9696±0.0008	0.7860±0.0050	0.9057±0.0002	372.76±9.76	2.46±0.08
SMAE	**0.8825±0.0028**	0.9703±0.0009	**0.7909±0.0048**	**0.9088±0.0002**	336.40±13.27	2.44±0.10

(a) (b) (c)

Fig. 8. (a) Pauli RGB image of dataset Oberpfaffenhofen. (b) Corresponding ground-truth map. (c) Terrains.

is superior to that of classification network based on SAE and SVM. Again, the validity of classification network based on the proposed SMAE and the correctness of the related analysis are proved.

E. Oberpfaffenhofen Image

This image is obtained from the Oberpfaffenhofen area in Germany, which provided by the German Aeropave Center's E-SAR with a size of 1300 × 1200. The Pauli RGB image and the ground-truth map are shown in Fig. 8. The scene is categorized into three basic classes: Built-up area, Wood land, and Open area. The total number of labeled pixels in the ground truth is 1 374 298. Considering this POL-SAR image to be a little large, 10% labeled data are randomly selected to calculate the classification accuracy table in [31]. However, we still use all the labeled pixels in all of our experiments. Another benefit for this operation is that the result image corresponding to the ground truth can be presented in a simple way.

As the previous analysis of parameter selection, the number of hidden layer nodes and sparsity parameter for AE, WAE, and the proposed MAE is set as 80 and 0.5 in this experiment, respectively. Here, we also change the parameter of softmax classifier to 1e−8, while other parameters are consistent with those of the Flevoland image. Besides, the network structure and parameters of SAE, SWAE, and SMAE are selected the same as those in the experiment of Flevoland. Including OA, the objective quantitative indexes for evaluating the performance of the different classification methods are listed in Table VIII and the maximum value of each of eight methods is marked in bold.

According to the data in Table VIII, it can be concluded that different classification methods have a better influence on different landforms. Wishart classifier has a lower correct rate for each category, especially Wood land and Built-up area. Although the classifier based on AE network, compared with the other two classifiers with the same structure, has a relatively good ability to recognize the pixels of Wood land, it cannot well process other two landscapes, particularly for pixels in Built-up area. On the contrary, classifiers based on WAE have the highest and lowest correct rates on the Built-up area and Wood land among three classifiers, respectively. In addition, the correct rate for each category of classifier based on the proposed MAE is somewhere in between. More importantly, the classifier based on the proposed MAE causes a small rise in OA compared with classifiers based on the AE and WAE. Furthermore, the OA of SVM is greater than that of the first four classification methods. The correct rate of each category has been improved correspondingly, except that of Built-up area. During the experiments, we also find the stability of classification network based on SAE is not as good as that of other classification networks. The correct rate of Wood land and Open area obtained by classification network based on SAE is the worst of all the methods, resulting in the OA of which is also the lowest. It occurs a phenomenon that the OA of classifier based on features acquired by the AE network is higher than that of the corresponding classification network. This shows the instability of the classification results when directly applying the AE network on the POL-SAR data matrix. On the other hand, the importance of considering data distribution into the design of AE is illustrated. Furthermore, the correct rate of Wood land and Built-up area produced by classification network based on SWAE and SMAE is much higher than that of the corresponding classifiers based on the feature extraction model. Simultaneously, the correct rate of Open area is very nearly the same. What is more, the OA of classification network based on the proposed SMAE is a bit higher than that of classification network based on SWAE. Equally, Wishart classifier does not require time to train. The training time of WAE is the longest of three classifiers based on feature extraction and it takes about 35 s. After completing the parameter optimization, SVM requires almost 65 s to accomplish the training process. It takes several hundred seconds to fulfill the training procedure of the three classification networks. Classification network based on the proposed SMAE costs about 336 s, which is the shortest among them. The time needed for achieving the prediction of the view image through Wishart classifier is no more than 1 min, while SVM has a longer predicting time of more than 1000 s. However, even more importantly, it only costs a few seconds for the remaining classification methods to accomplish the prediction of millions of pixels.

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TABLE IX
CLASSIFICATION PERFORMANCE ON THE SAN FRANCISCO BAY IMAGE OF TWO PROPOSED METHODS AND SIX COMPARING METHODS

	Water	Vegetation	High-density urban	Low-density urban	Developed urban	OA	$T_{training}$ (s)	$T_{predicting}$ (s)
Wishart	0.9705±0.0033	**0.9138±0.0158**	0.5199±0.0471	0.7035±0.0538	0.6324±0.0825	0.8275±0.0019	0.00±0.00	93.97±0.62
AE	0.9976±0.0016	0.8358±0.0464	0.6984±0.0260	0.7689±0.0305	0.5759±0.0446	0.8705±0.0017	10.20±1.87	2.19±0.07
WAE	0.9961±0.0015	0.8660±0.0106	0.6755±0.0080	0.8084±0.0046	0.6353±0.0205	0.8761±0.0013	54.87±1.23	1.83±0.05
MAE	0.9981±0.0017	0.8466±0.0441	0.7060±0.0488	0.7740±0.0490	0.6328±0.0468	0.8810±0.0024	50.59±17.55	2.12±0.14
SVM	0.9965±0.0012	0.8897±0.0025	0.7435±0.0074	0.8232±0.0029	0.6412±0.0069	0.8934±0.0006	59.84±2.94	**2174.72±76.17**
SAE	0.9014±0.2149	0.8629±0.0410	0.7209±0.0402	0.7755±0.0397	0.6530±0.1646	0.8768±0.0048	**564.64±127.87**	7.90±0.13
SWAE	0.9988±0.0001	0.8823±0.0016	0.7632±0.0034	0.8127±0.0029	0.7319±0.0041	0.8984±0.0001	523.77±8.44	4.01±0.09
SMAE	**0.9995±0.0000**	0.8862±0.0013	**0.7805±0.0026**	**0.8306±0.0022**	**0.7355±0.0039**	**0.9056±0.0000**	466.68±8.09	3.97±0.07

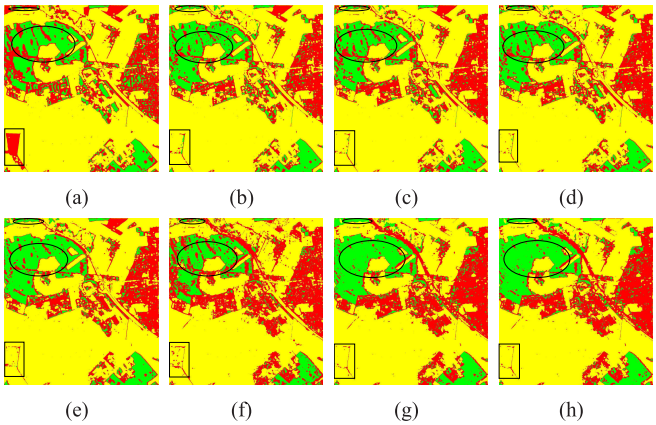

Fig. 9. Prediction result images of the whole Oberpfaffenhofen image generated by different classification methods. (a) Wishart. (b) AE. (c) WAE. (d) MAE. (e) SVM. (f) SAE. (g) SWAE. (h) SMAE.

Fig. 9 shows the prediction result images of different methods. There is an obvious case of wrong classification in the result image of Wishart classifier, which marked with a black rectangle in Fig. 9(a). Most of the Open area pixels are mistaken for Built-up area. The reason may be that the ground-truth map of this image contains more labeled data; 5% labeled data selected as the training set is not a good choice to the Wishart classifier. However, only a few samples are misplaced in the result images of Fig. 9(b)–(h). According to the corresponding ground-truth map, the bottom-left corner of the image belongs to the Built-up area. As shown in Fig. 9(a)–(h), Wishart classifier and classification network based on SAE have the ability to recognize most of the pixels correctly, while most of the pixels are falsely classified to Wood land in other methods. We can from the black ellipses in Fig. 9(a)–(d), which mainly used to represent the Wood land, find that the classification result images of the three classifiers based on the feature extraction are smoother than those of Wishart classifier. In view of the whole image, it is difficult to distinguish the best result image from Fig. 9(b)–(d) with the naked eyes, which verified the previous analysis about the correct rate of corresponding three classifiers. Fig. 9(e) shows the result image of SVM, which looks neat, and it proves the powerful classification ability of SVM, while the result image of classification network based on SAE shown in Fig. 9(f) is worse than that of the corresponding classifier based on feature extraction, which shows that using the labeled data to train the process of AE may be not a good choice.

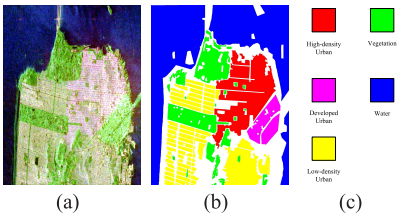

Fig. 10. (a) Pauli RGB image of dataset San Francisco Bay. (b) Corresponding ground-truth map. (c) Terrains.

The instability of classification network based on SAE is verified again. The result images given in Fig. 9(g) and (h) are evidently cleaner and smoother than the result image of the corresponding classifiers based on feature extraction shown in Fig. 9(c) and (d). The validity of classification network structure is proven again. In addition, Fig. 9(h) looks better than Fig. 9(g), which explained the effectiveness of classification network based on the proposed SMAE.

F. San Francisco Bay Image

The RADARSAT-2 C-band data of San Francisco Bay are used as the last experimental image. The size of the selected scene is 1380 × 1800, which contained by mainly five classes: High-density urban, Water, Vegetation, Developed, and Low-density urban. The Pauli RGB image and the ground-truth map are given in Fig. 10; 1 804 087 is the total number of labeled pixels in the ground truth.

According to the previous analysis of the parameter selection, we select 80 as the number of hidden layer nodes and apply 0.05 as sparsity parameter in this experiment. In the same way, the weight of weight decay term and sparsity penalty term are made use of 1e−8 and 3, respectively. Moreover, the parameter of softmax classifier is set as the default value. Parameters of three classification networks for this data set are chosen the same as those of the Flevoland experiment. At last, the objective quantitative indexes for evaluating the performance of the different classification methods for San Francisco Bay image are given in Table IX.

The following conclusions are drawn from the data in Table IX. Except for Wishart classifier and classification network based on SAE, the correct rate of all the other classification methods for Water can reach more than 99%. Wishart classifier has the highest correct rate of the Vegetation. However, the correct rate for three different urban areas

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obtained by Wishart classifier is almost the lowest among all the compared classification methods. The correct rate of three categories obtained by classifier based on the proposed MAE is the highest in the three classifiers based on feature extraction. For three classifiers based on different feature extraction networks, the classification correct rate of classifier based on AE network for three urban areas is either lower than that of classifier based on WAE model or lower than that of classifier based on the proposed MAE. Moreover, the correct rate of two classifiers based on WAE and MAE networks for the High-density urban and Low-density urban is more than 3% higher than that of each other. For Developed urban, there is no much difference between the correct rate of two classifiers based on WAE and MAE network. To summarize, the OA of classifier based on the proposed MAE is the highest for the first four classifiers. Besides, the classifier based on WAE rank second with a gap of only 0.4% in classification OA. The correct rate of most of the categories obtained by SVM is nearly the same as those of the classification network based on SWAE and is higher than those of classification network based on SAE. For all categories, the correct rate of classification network based on the proposed SMAE is the highest among the three classification networks. The OA of three classification networks is at least 1% higher than that of the corresponding classifiers based on the feature extraction. Generally, the training time of classification network based on SAE is the longest and costs about 564 s, while SVM takes nearly 2174 s to predict all the pixels in the whole image, which is the longest of all the methods. The training time spent on the three classifiers based on different feature extraction networks and the corresponding classification networks is one and two orders of the magnitude, respectively. However, the predicting time spent on three classifiers based on different AE networks and the corresponding classification networks with different stacked AE networks has the same order of magnitude. More specifically, the time spent on the training process of classifier based on WAE and classification network based on SAE is the longest in the three classifiers based on the feature extraction and the corresponding classification networks, respectively. It takes a few seconds to finish the predicting process of the other classification methods, besides Wishart classifier and SVM.

We can also make the following conclusions from the prediction result images shown in Fig. 11. All the result images have a good classification result to Water, except Fig. 11(a). For the predicting images shown in Fig. 11(a)–(d), it seems that the four methods have the same classification performance on Developed urban. From the result images in Fig. 11(b)–(d), it is easy to find that the corresponding three classifiers cannot distinguish High-density urban from Low-density urban very well. According to the areas marked with the oval frame, it can clearly observe that the best result for classification of the High-density urban is produced by classifier based on the proposed MAE. Yet, it is hard to tell which result image is the best for Low-density urban intuitively. These regions framed by the black rectangles in the image Fig. 11(c) look very messy. It might imply that classifier based on WAE mistakenly classifies a large number of pixels in the Vegetation as Water.

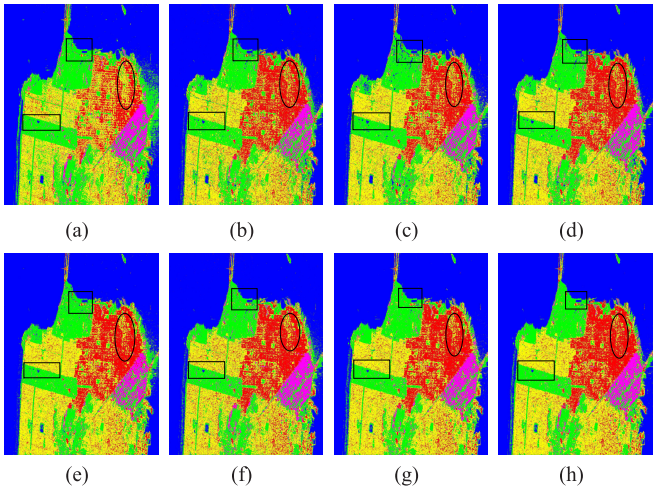

Fig. 11. Prediction result images of the whole San Francisco Bay image generated by different classification methods. (a) Wishart. (b) AE. (c) WAE. (d) MAE. (e) SVM. (f) SAE. (g) SWAE. (h) SMAE.

This disorderly phenomenon has been obviously suppressed in the corresponding regions of the images in Fig. 11(b) and (d), and the validity of classifier based on the proposed MAE is proven again. Fig. 11(e) shows the result image of SVM, in which the classification effect for each category seems to be very good. The marked areas in Fig. 11(f)–(h) are obviously smoother than the corresponding areas of Fig. 11(b)–(d), which shows the effectiveness of the multilayer classification network. In addition, the marked areas of Fig. 11(h) are also better than those of Fig. 11(f) and (g), which illustrated the advantage of the proposed method. In a word, the observation confirms the results of data analysis once more.

G. Discussion on the Random Initialization of Parameters and the Ratio of Training Data

During the experiment, the OA will be affected by the random selection of the initial parameters. Thus, we run three classifiers based on feature extraction 30 times on three testing images and calculate their means. The results for each testing image are shown in Fig. 12. The three colors of RGB represent the results of AE, WAE, and the proposed MAE method, respectively. The solid line corresponds to the OA of each iteration, while the broken line represents the average correct rate. We can draw the following conclusions from Fig. 12. Classifier based on the proposed MAE has a certain improvement in OA. Compared with other two classifiers, the OA curve of classifier based on AE appears to have the fluctuation with different amplitudes, which stated that the stability of classifier based on AE is not as good as that of other classifiers.

Fig. 13 shows the classification results of classifier based on the proposed MAE and classification network based on SMAE in terms of OA, which is regarded as a function of the number of training samples per class. From Fig. 13(a) and (c), we can see that as the proportion of training data increases, the OA of two classification methods increases and tends to be stable after the ratio reaches a critical point. By this method, we can

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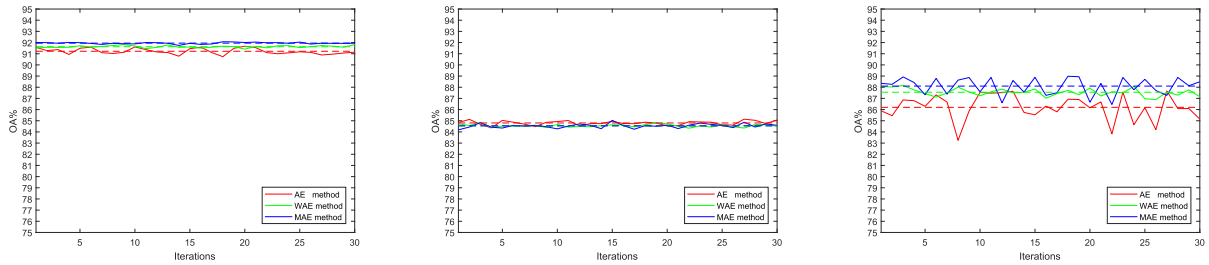
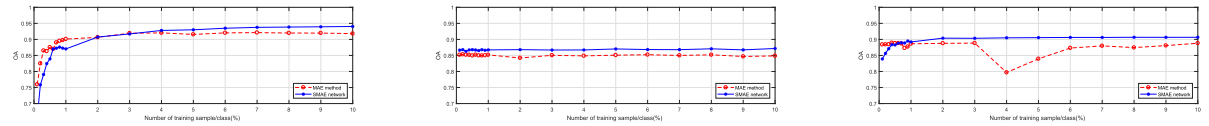


(a) (b) (c)

Fig. 12. OA curve of three methods with 30 iterations on different images. (a) Flevoland. (b) Oberpfaffenhofen. (c) San Francisco Bay.



(a) (b) (c)

Fig. 13. OA of the proposed MAE and SMAE used as the function of the number of training samples over three POL-SAR images. (a) Flevoland. (b) Oberpfaffenhofen. (c) San Francisco Bay.

make sure the minimum amount of training data required for each algorithm to achieve a satisfactory result. It can also be found that classifier based on the proposed MAE is higher in OA than that of classification network based on the proposed SMAE when the proportion of training data is small. In other words, the proposed MAE has a better ability to obtain the features needed for the classifier in this case. For classification network based on the proposed SMAE, the ratio of training data per class may be too less to satisfy the requirement of the four-layer classification network. With the ratio of training data increasing, this situation is reversed. From Fig. 13(b), we can also find that 1/1000 of the amount of the training data is enough to the two proposed classification methods. This is due to the number of labeled pixels for per class which are too large in the corresponding ground-truth image. Moreover, the OA of classification network based on the proposed SMAE is always higher than that of classifier based on the proposed MAE, which proved the validity of the multilayer classification network.

V. Conclusion and Future Work

A more detailed and rigorous analysis on the distribution of POL-SAR data matrix, rearranged by the coherent matrix, is given at first. A novel data normalization method is also put forward in succession. Combined with the data distribution of different rows in the POL-SAR data matrix, an unsupervised feature learning method is presented through solving a new energy function in the process of AE network. Then, a new classification method for POL-SAR image is accomplished by means of a softmax classifier in this article. Considering the data distribution of the input layer and the consistency of data distribution between the output of the former layer and the input of the adjacent layer, a new modified stacked AE network is also presented for mapping the original input vector into a higher layer feature space. After that, a specific four-layer discriminant classification network is proposed in this article by discarding the decoding process of the two hidden layer stacked AE networks and connecting with a softmax classifier. Three real POL-SAR data sets are used as the experiments. A discussion of parameter selection for the proposed MAE and two related comparison methods has been made as well. As well, it is generalized to the corresponding classification network. Including the correct rate of each category, this article gives some impersonal indicators to measure the classification performance and shows the prediction result image at the same time. Compared with one classical and two related methods, the results of both the proposed methods show better agreement with the corresponding ground truth, a certain increase in OA as well as greater ability to predict the class of the pixels in the whole image. At last, we also provide a debate on the ratio of training data and the random initialization of parameters.

The future work can be carried out with the following aspects. The neighborhood information is added while taking the distribution of a different form of POL-SAR data into consideration in the design of the feature extraction model. As one of the postprocessing methods, spatial location information is exploited to repress the noisy phenomenon of misclassified pixels in the classification result image. We can also make a qualitative analysis ahead of time. To be specific, the confusion matrix is used to determine the main category of misclassification for each category, and then, the spatial location information is combined to obtain a better classification prediction result image.

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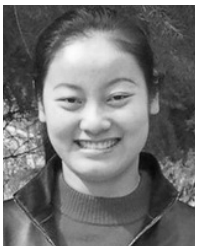

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